DORSA CHARKHIAN

Human-AI Interaction · Inclusive & Participatory Design · Digital Health

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RESEARCH INTERESTS

Interactive machine learning and personal sensing for digital health; participatory and systemic design of AI systems; trust, accessibility and cultural adaptation of AI assistants for multilingual and immigrant communities.

EDUCATION

M.S. in Artificial Intelligence, Touro University, New York *Expected Fall 2026*

M.S. in Digital Media (User Experience & Interaction Design), Drexel University, Philadelphia *June 2025*

Thesis: Exploring the Potential of AI for Improving Inclusivity in University Portals. Advisor: Prof. Troy Finamore

B.F.A. in Islamic Art – Wood, Tabriz Art University, Iran *June 2022*

PUBLICATIONS

Full list and citations: Google Scholar profile

Peer-Reviewed Conference Papers — Human-AI Interaction & Design

- **Charkhian, D.**, & Moghaddami, B. (2025). *How can AI evaluate and improve inclusivity in university portals, with a focus on cultural, linguistic, and accessible requirements?* INTED2025 Proceedings.
- **Charkhian, D.**, & Moghaddami, B. (2025). *Making university portals more inclusive: A design-focused approach to AI integration*. EDULEARN25 Proceedings, 242–247.
- Saebi, P., & **Charkhian, D.** (2026). *Human–AI co-design in visual communication: Ethical decision-making in AI-assisted ideation*. INTED2026 Proceedings.
- Saebi, P., **Charkhian, D.**, & Holohan, K. (2026). *From solo designer to co-creator: Reconstructing design thinking frameworks for human-AI collaborative ideation*. EDULEARN26 Proceedings.
- Saebi, P., **Charkhian, D.**, & Holohan, K. (2026). *Redefining design expertise: Competency markers and authentic creativity in the age of generative AI*. EDULEARN26 Proceedings.
- Bahrami, S., & **Charkhian, D.** (2025). *AI chatbots and speculative futures: A comparative study of online-asynchronous and in-person digital media education*. EDULEARN25 Proceedings.

Journal Articles — AI & Machine Learning

- Mohammadabadi, S. M. S., Moghaddam, A. K., Entezami, M., Seyedrezaei, M., **Charkhian, D.**, Moghaddami, B., & Sassani, M. (2026). *Advancing electric vehicle ecosystems: A survey of generative artificial intelligence and distributed machine learning applications*. Global Energy Interconnection (Elsevier).
- Entezami, M., Basirat, S., Moghaddami, B., Bazmandeh, D., & **Charkhian, D.** (2025). *Examining the importance of AI-based criteria in the development of the digital economy: A multi-criteria decision-making approach*. Journal of Soft Computing and Decision Analytics, 3(1), 72–95.
- Taghizadeh, A., Karaminezhad, K., Fakhri, N., Moghaddami, B., & **Charkhian, D.** (2025). *Assessing innovation strategies in the digital economy through artificial intelligence-based criteria using CoCoSo method*. Journal of Intelligent Decision Making and Granular Computing, 1(1), 237–255.

Thesis & Book Chapter

- **Charkhian, D.** (2025). *Exploring the potential of AI for improving inclusivity in university portals* [Master's thesis, Drexel University]. https://doi.org/10.17918/00011005
- **Charkhian, D.** (2025). *Feminism, media and technology: Navigating the digital landscape*. In MAD (Media, Arts and Design) Anthology III. Drexel University.

Research Contributions

- Research contributor (non-author), user-flow research: *Plumes and Pixels: Bridging digital and traditional art in SciArt education*, ACM SIGGRAPH Asia 2024 Educator's Forum.

RESEARCH EXPERIENCE

Lead Researcher – Multilingual AI Assistant for Inclusive University Portals *Jul 2024 – Jun 2025*

Drexel University · Funded by IDEA-DEC MicroGrant · Advisor: Prof. Troy Finamore

- Mixed-methods study of barriers faced by international students in the DrexelOne portal: surveys (n=34), semi-structured interviews (n=9), IT support log analysis and task-based usability testing (n=15).
- Designed and evaluated D.A.V.E., a multilingual AI assistant with adaptive navigation; improved task success and reduced stress for Farsi-, Spanish- and Chinese-speaking students.
- Found that users preferred culturally appropriate language over "robotic" machine translation, identifying dialect support as a key limitation for trust.
- Led a funded two-person team (UI designer, web developer), including hiring and project direction.

Researcher – Speculative Design of Sleep-Tracking Technologies
Drexel University · Graduate research project with Prof. John Seberger

- Explored the ethics of wearable health sensing, focusing on privacy, trust and user autonomy; designed speculative scenarios balancing data-driven insight with user control.

UX Researcher & Service Designer – Supporting Parents in Childcare
Drexel University · Academic team project · Participatory service design

- Co-designed community-centered health solutions, including wearable devices and mobile clinics, to support children's health.

UX Researcher – VR Fashion Education *Jul 2025 – May 2026*
Drexel University · IRB-certified (CITI)

- Designed and ran surveys and interviews with 50+ students across 3 courses; iterative testing increased engagement by 45% and comprehension by 50%.

SELECTED TECHNICAL PROJECTS

Applied AI Agents & Automation (client projects)

Multi-Agent AI Trading System – confidential client, small investment firm *Jun 2026 – Present*

- Designed an AI decision-support system that gives the client daily, research-backed trade recommendations (entry, stop-loss, target, risk/reward, confidence) while the client keeps final approval over every trade.
- Iterated from a six-role design to a final two-agent build with hard risk rules, a trade journal and persistent memory of lessons learned; tested through paper trading before any real use.

Inventory Automation & Digital Archive – client, Persian rug business *May – Jul 2026*

- Digitized a 700+ item inventory: reconciled paper records with photos by hand, then connected Claude to Notion so records were entered into a searchable database, and used Claude to auto-generate illustrated catalogs.

Interactive Sensing

TIME – interactive 360° environment (TouchDesigner) *2024*

- Real-time webcam head tracking lets viewers navigate a panorama of homes across three cities with no controller; adaptive generative soundtrack. Video

INTERACT – body-responsive media installation (TouchDesigner) *2024*

- Webcam-based body-movement sensing controls an archive of images of Iranian women. Video

GRANTS & AWARDS

IDEA-DEC MicroGrant, Westphal DEI Council, Drexel University *Nov 2024*
Competitive award funding AI-driven research on inclusive university portals for international students.

COMAD General Award (RTGT), College of Media Arts & Design, Drexel University *2024*

Third Place, PhillyCHI Company of SE UX Competition *2024*

TEACHING EXPERIENCE

Adjunct Instructor, Department of Technology, New York University *Mar – May 2026*

- Taught technology courses to 20 students across two classes.

Visiting Assistant Professor, School of Information, Pratt Institute *Aug 2025 – Jan 2026*

- Taught graduate UX and AR/VR (Unity, A-Frame, XR usability testing); mentored student projects on accessibility and inclusion.

Teaching Assistant, User Experience Design, Drexel University *Sep 2023 – Jun 2025*

PROFESSIONAL EXPERIENCE

UX Design & Research Fellow, Trops (Colla Netrix, Inc.) *Sep 2025 – Mar 2026*

- Led user research and usability testing for a community-centered mobile app; reported to the CTO.

Site Coach, Liberty Partnerships Program, Touro University *Jul – Aug 2026*

- Mentored middle and high school students on academic and career planning.

UX Researcher, Academy of Natural Sciences of Drexel University *Feb – Jun 2023*

- Researched user flow and content structure for an interactive 3D digital twin of museum headwear (Plumes and Pixels, later presented at ACM SIGGRAPH Asia 2024 Educator's Forum).

SERVICE & LEADERSHIP

Ambassador, Google Women Techmakers *2026 – Present*

Selected for the 2026 global cohort; community focus: AI agents in educational systems.

Peer Review

- **Distinguished Reviewer Board,** ACM Transactions on Computer–Human Interaction (TOCHI), 2026–
- **ACM CHI 2026** Posters Track reviewer, incl. a home-based virtual healthcare agent for older adults
- **International Journal of Human–Computer Interaction** (Taylor & Francis), 2026: generative-AI ideation support
- **AI and Ethics** (Springer), 2026: AI-enabled home and body sensing (e.g., continuous glucose monitors), surveillance and care
- **AI & Society** (Springer), 2025; **International Journal of Educational Technology in Higher Education** (Springer), 2025–2026
- **UXPA International 2026 Conference,** Chair Reviewer
- **Thomas Jefferson University,** external reviewer, M.S. UX & Interaction Design thesis presentations, 2025

Mentoring & Community

- **Mentor, AI Ethics & Inclusive UX,** DIFC Innovation Hub – Ignyte (Dubai), 2025 – Present
- Mentor, PhillyCHI × AIGA Philly UX Mentorship Program (2025); ADPList Fellow Mentor

SKILLS

Research methods: mixed methods, interviews, surveys, usability and task-based testing, participatory and speculative design, thematic analysis (ATLAS.ti, MAXQDA, NVivo), IRB/CITI certified

AI systems & programming: AI agent and workflow design, prompt engineering, AI-assisted automation (Claude, MCP integrations, Notion)

Interactive & ML prototyping: TouchDesigner (webcam head/body tracking), Teachable Machine, ComfyUI/Stable Diffusion, Unity, A-Frame, HTML/CSS/JavaScript

Design: Figma, Miro, Photoshop, Rhino, MatrixGold

Training: Anthropic Academy: Claude 101, Claude Code 101 (2026); Google 5-Day AI Agents Intensive; CITI human-subjects research certification

Languages: English, Farsi